PhD report for the academic year 2017/2018

Weak solution of the merged mathematical equations of the polluted atmosphere

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The main ideas of the model

- Geophysical fluid $\rightarrow$ shallow domain characteristics — the domain is greater horizontally than in the vertical dimension.
- Classical / general concepts as foundation:
  - The wind flow being modelled by the incompressible Navier-Stokes equations
  - Using a convection - diffusion equation to describe the pollution effect
- Relatively new or more specific ideas:
  - Taking advantage of a downwind-direction matching coordinate system
  - Using a Gaussian pulse as a source term
  - Rescaling the diffusivity parameters according to the largely different horizontal and vertical scales

The governing equations

- Fluid velocity equations — incompressible Navier-Stokes equations with the density taken to be identically equal to one.

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v} - \Delta_\nu \mathbf{v} + \nabla q + 2\mathbf{w} \times \mathbf{v} = \mathbf{g} \text{ in } \Omega_\epsilon \times (0, T)$$

$$\nabla \cdot \mathbf{v} = 0 \text{ in } \Omega_\epsilon \times (0, T)$$

- A convection-diffusion equation representing pollutants in the atmosphere:

$$\partial_t C + \mathbf{v} \cdot \nabla C = K_x \partial^2_{xx} C + K_y \partial^2_{yy} C + K_z \partial^2_{zz} C + S$$

Scaling

One of the main tools to exploit the "almost" 2D structure of the atmosphere → introduce the aspect ratio

$$\epsilon = \frac{\text{characteristic depth}}{\text{characteristic width}},$$

which incorporates the shallowness in a mathematical way. Natural scaling process:

1. $x = x_1, y = x_2, z = \epsilon x_3$
2. $v_1 = u_1^\epsilon, v_2 = u_2^\epsilon, v_3 = \epsilon u_3^\epsilon, p = p^\epsilon,$
3. $\nu_x = \nu_1, \nu_y = \nu_2, \nu_z = \epsilon^2 \nu_3,$
4. $K_y = K_2, K_z = \epsilon^2 K_3.$

Scaling - Part 2

Additional scaling equation: different in nature from the previous ones $\rightarrow$ it is not derived from the shallowness of the domain — it is suggested by the fact that in the downwind direction it is natural to assume that *the diffusion is negligible compared to downwind transport* i.e.

$$\left|u_1 \partial_{x_1} C\right| \gg \left|K_x \frac{\partial^2 C}{\partial x_1^2}\right|,$$

which leads us to using

$$K_x = \epsilon K_1.$$

Local domain and boundary conditions

We use a local cartesian coordinate system so that we can neglect the curvature of the Earth.

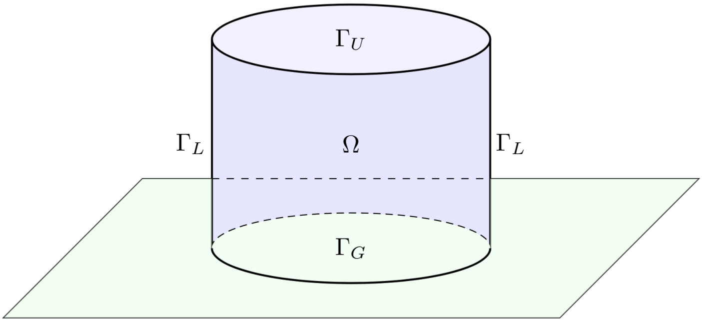

Figure: A local slice of the atmosphere

Source term

For modelling the source function of the concentration we use the Gaussian impulse approach.
We use a parametrised nascent delta function to describe the source term, with the parameter being ϵ.
We have

$$S_\epsilon(t, t_s, x, x_s) = s(t, t_s)\delta_\epsilon(x - x_s),$$

with the activation term $s(t, t_s) = IH(t - t_s)$ where $H(\cdot)$ is the Heaviside step function. In the limit model we naturally arrive to the delta distribution as a source.

Anisotropic equations

In $\Omega \times (0, T)$ we have the following system.

$$\partial_t u_1^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_1^\epsilon - \Delta_\nu u_1^\epsilon - \alpha u_2^\epsilon + \epsilon \beta u_3^\epsilon + \partial_1 p^\epsilon = 0$$

$$\partial_t u_2^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_2^\epsilon - \Delta_\nu u_2^\epsilon + \alpha u_1^\epsilon + \partial_2 p^\epsilon = 0$$

$$\epsilon^2 \{\partial_t u_3^\epsilon + \mathbf{u}^\epsilon \cdot \nabla u_3^\epsilon - \Delta_\nu u_3^\epsilon\} - \epsilon \beta u_1^\epsilon + \partial_3 p^\epsilon = 0$$

$$\nabla \cdot \mathbf{u}^\epsilon = 0$$

$$\partial_t C^\epsilon + \mathbf{u}^\epsilon \cdot \nabla C^\epsilon = \epsilon K_1 \frac{\partial^2 C^\epsilon}{\partial x_1^2} + K_2 \frac{\partial^2 C^\epsilon}{\partial x_2^2} + K_3 \frac{\partial^2 C^\epsilon}{\partial x_3^2} + S_\epsilon$$

Anisotropic equations - Part 2

The initial and boundary conditions are chosen to be

$$\mathbf{u}^{\epsilon} = 0 \text{ on } \Gamma \times (0, T),$$

$$\mathbf{u}^{\epsilon}(\cdot, t = 0) = \mathbf{u}_0, C^{\epsilon}(\cdot, t = 0) = C_0^{\epsilon} \text{ in } \Omega,$$

$$C^{\epsilon} = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C^{\epsilon} = \partial_3 C^{\epsilon} = 0 \text{ on } \Gamma_G \times (0, T),$$

$$\partial_1 C^{\epsilon} \in L^2 L^2(0, T; \Gamma_G).$$

Hydrostatic equations

$$\partial_t u_1 + \mathbf{u} \cdot \nabla u_1 - \Delta_\nu u_1 - \alpha u_2 + \partial_1 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t u_2 + \mathbf{u} \cdot \nabla u_2 - \Delta_\nu u_2 - \alpha u_1 + \partial_2 p = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_3 p = 0 \text{ in } \Omega \times (0, T)$$

$$\nabla \cdot \mathbf{u} = 0 \text{ in } \Omega \times (0, T)$$

$$\partial_t C + \mathbf{u} \cdot \nabla C = K_2 \frac{\partial^2 C}{\partial x_2^2} + K_3 \frac{\partial^2 C}{\partial x_3^2} + S \text{ in } \Omega \times (0, T)$$

Hydrostatic equations - Part 2

$$u_1 = u_2 = u_3 n_3 = 0 \text{ on } \Gamma \times (0, T)$$

$$u_i(\cdot, t = 0) = u_{0i}, C(\cdot, t = 0) = C_0 \text{ in } \Omega, i = 1{,}2$$

$$C = 0 \text{ on } (\Gamma_U \cup \Gamma_L) \times (0, T), \partial_2 C = \partial_3 C = 0 \text{ on } \Gamma_G \times (0, T),$$

$$\partial_1 C \in L^2 L^2(0, T; \Gamma_G)$$

Main theorem

We show a rigorous connection between the two sets — taking the limit in the solutions we arrive to a justification of the hydrostatic model for the case of the polluted atmosphere.

Theorem

Let $\mathbf{u}_0 \in L^2(\Omega)^3$, with $\nabla \cdot \mathbf{u}_0 = 0, \mathbf{u}_0 \cdot \mathbf{n} = 0$ on $\partial\Omega$; and let $C(0) \in L^2(\Omega)^3$, $C(0) = 0$ on $\partial\Omega$; then as the aspect ratio ϵ tends to zero, any weak solution $(\mathbf{u}^\epsilon, C^\epsilon)$ of the anisotropic equations converge to a weak solution $(\mathbf{u},\ C)$ of the hydrostatic equations of the polluted atmosphere.

ModCompShock

Energy inequality

$$
\begin{aligned}
&\frac{1}{2}\Big(\big\|\mathbf{u}_H^\epsilon(t)\big\|_{L^2}^2 + \epsilon^2\big\|u_3^\epsilon(t)\big\|_{L^2}^2 + \big\|C^\epsilon(t)\big\|_{L^2}^2\Big) + \int_0^t \left[\|\nabla_\nu \mathbf{u}_H^\epsilon\|_{L^2}^2 + \epsilon^2\|\nabla_\nu u_3^\epsilon\|_{L^2}^2\right] \mathrm{d}\tau \\
&+ \int_0^t \Bigg[\Big(\epsilon K_1 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1)\Big)\|\partial_1 C^\epsilon\|_{L^2}^2 + \Big(K_2 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1)\Big)\|\partial_2 C^\epsilon\|_{L^2}^2 \\
&+ \Big(K_3 - \zeta\frac{\epsilon}{2}K_1 c_T(c_S+1)\Big)\|\partial_3 C^\epsilon\|_{L^2}^2\Bigg] \mathrm{d}\tau \\
&\leq \frac{1}{\zeta}K_1\frac{\epsilon}{2}\int_0^t \|\partial_1 C^\epsilon\|_{L^2(\Gamma_G)}^2 + \frac{1}{2}\Big(\big\|\mathbf{u}_H^\epsilon(0)\big\|_{L^2}^2 + \epsilon^2\big\|u_3^\epsilon(0)\big\|_{L^2}^2 + \big\|C^\epsilon(0)\big\|_{L^2}^2\Big) \\
&+ \int_0^t (S^\epsilon, C^\epsilon)\,\mathrm{d}\tau
\end{aligned}
$$

Proof steps - Part 1

1. The beginning step of the proof is to obtain the energy inequality that yields a priori estimates such as $L_t^\infty L_x^2$ boundedness for C^ϵ; $L_t^2 L_x^2$ boundedness for $\sqrt{\epsilon}\partial_1 C^\epsilon, \partial_2 C^\epsilon, \partial_3 C^\epsilon$.
2. Once we have boundedness, we have the weak convergence of the linear terms, while the weak convergence of the nonlinear terms not including the pollution term had been shown previously by Azérad and Guillén (SIAM Journal on Mathematical Analysis, 2001).
3. We prove the convergence of the term $(\mathbf{u}^\epsilon C^\epsilon, \nabla \tilde{C})$ by using a compactness result. It can be proved that $u_i^\epsilon C^\epsilon \rightharpoonup u_i C$ following these two steps:

Proof steps - Part 2

- we have a bound for the time derivative of the concentration:

$$\frac{\partial C^\epsilon}{\partial t} = -u^\epsilon \nabla C^\epsilon + \sqrt{\epsilon} K_1 \partial_1 \partial_1 \sqrt{\epsilon} C^\epsilon + K_2 \partial_2 \partial_2 C^\epsilon + K_3 \partial_3 \partial_3 C^\epsilon - S^\epsilon,$$

  Eventually we have $\frac{\partial C}{\partial t} \in L^1 W^{-1,1}$,
- to prove $\left\| u_i^\epsilon - u_i^\epsilon(\cdot + \xi, t) \right\|_{L^2 L^2} \to 0$ as $\xi \to 0$, uniformly in ϵ.

Combining these results allow us to apply a compactness theorem that yields the convergence.

Outlook

Directions and ideas for future work:

1. Use a diffusion matrix instead of the original coefficients
2. Investigate the case of strong convergence
3. Visualise the results using the finite element method

University of Sussex - Prof. Omar Lakkis

The main results and steps are the following:

1. Software packages such as Docker, Anaconda, ParaView, FEniCS and Alberta have been installed
2. Writing scripts to login / connect to the HPC server at the Sussex University
3. Getting to know FEniCS: tutorial files and the FEniCS version of the "hello world" programs
4. Creating a basis for future work with specific theoretical details that arise when implementing this project - such as the loss of incompressibility when it comes to the FEM approach, and the use of Lagrange multipliers.

Lectures

1. Numerical methods for hyperbolic problems - Prof. G. Russo (University of Catania), GSSI
2. Physics of the atmosphere - Prof G. Curci, Prof. G. Redaelli, University of L'Aquila

Seminars, workshops, summer schools

1. Intensive Program on Fluids and Waves, GSSI
2. Women in applied and computational Mathematics, GSSI — **The topic and the results of this present research project were represented - together with the work of many early-stage female researchers - in the poster session.**
3. Evans function methods in the stability analysis of periodic wavetrains, Prof. R. Plaza
4. On the spectral stability of traveling fronts for reaction diffusion-degenerate equations, Prof. R. Plaza
5. Waves in Flows Summer School in Prague